Underwater Sound Analysis Report

Baltic Sea Acoustic Clips (Doctoral Applicants Assignment)

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1 Executive Summary

This report presents a complete acoustic analysis of six underwater sound clips extracted from a Baltic Sea recording. The workflow includes:

- Time-domain waveform analysis,
- Spectrogram-based time–frequency characterization,
- Broadband level estimation in dB re microPa,
- One-third octave band level analysis,
- Feature extraction (RMS, peak, crest factor, dominant frequency, spectral centroid, zero-crossing rate),
- Interpretation of likely source characteristics.

2 Data Overview

- Source file: `untitled.flac`
- Sampling rate: 16 000 Hz
- Total duration: 57.52 s
- Analysis region: 20–28 s (main acoustic event)
- Number of clips: 6 clips, each 1 s
- Acoustic reference: 1 µPa (underwater acoustics standard)

Clip time windows:

1. Clip 1: 20.0–21.0 s
2. Clip 2: 21.4–22.4 s
3. Clip 3: 22.8–23.8 s
4. Clip 4: 24.2–25.2 s
5. Clip 5: 25.6–26.6 s
6. Clip 6: 27.0–28.0 s

3 Methodology

3.1 Signal Processing Pipeline

The recording was scanned for energetic events. The dominant event spans approximately 20–28 s, from which six non-overlapping 1 s clips were extracted for consistent comparison.

3.2 Broadband Sound Level

Broadband level is computed as:

$$L_B = 20 \log_{10} \left(\frac{p_{\mathrm{RMS}}}{p_{\mathrm{ref}}} \right), \tag{1}$$

where $p_{\mathrm{ref}} = 1\,\mu\mathrm{Pa}$.

3.3 One-Third Octave Band Analysis

- Standard center frequencies from 63 Hz to 10 000 Hz.
- 4th-order Butterworth band-pass filters.
- Band levels computed in dB re microPa.

3.4 Spectral Analysis

- STFT spectrograms with logarithmic frequency axis.
- Welch PSD estimate (4096-point FFT).

3.5 Extracted Features

- RMS amplitude (V),
- Peak amplitude (V),
- Crest factor (dimensionless),
- Dominant frequency (Hz),
- Spectral centroid (Hz),
- Zero-crossing rate (dimensionless per frame).

4 Results

4.1 Broadband Levels and Spectral Indicators

Table 1: Key measurements for each clip

Clip	Time Window (s)	Broadband Level (dB re 1 µPa)	Dominant Freq. (Hz)	Spectral Centroid (Hz)
1	20.0–21.0	85.4	53	102.7
2	21.4–22.4	85.2	54	97.0
3	22.8–23.8	87.3	53	79.6
4	24.2–25.2	87.7	53	78.9
5	25.6–26.6	86.3	53	86.8
6	27.0–28.0	85.6	53	90.0

Aggregate summary:

- Average broadband level: 86.3 dB re microPa
- Level range: 85.2–87.7 dB re microPa
- Dominant frequency is consistently around 53 Hz

5 Figures

5.1 Waveforms

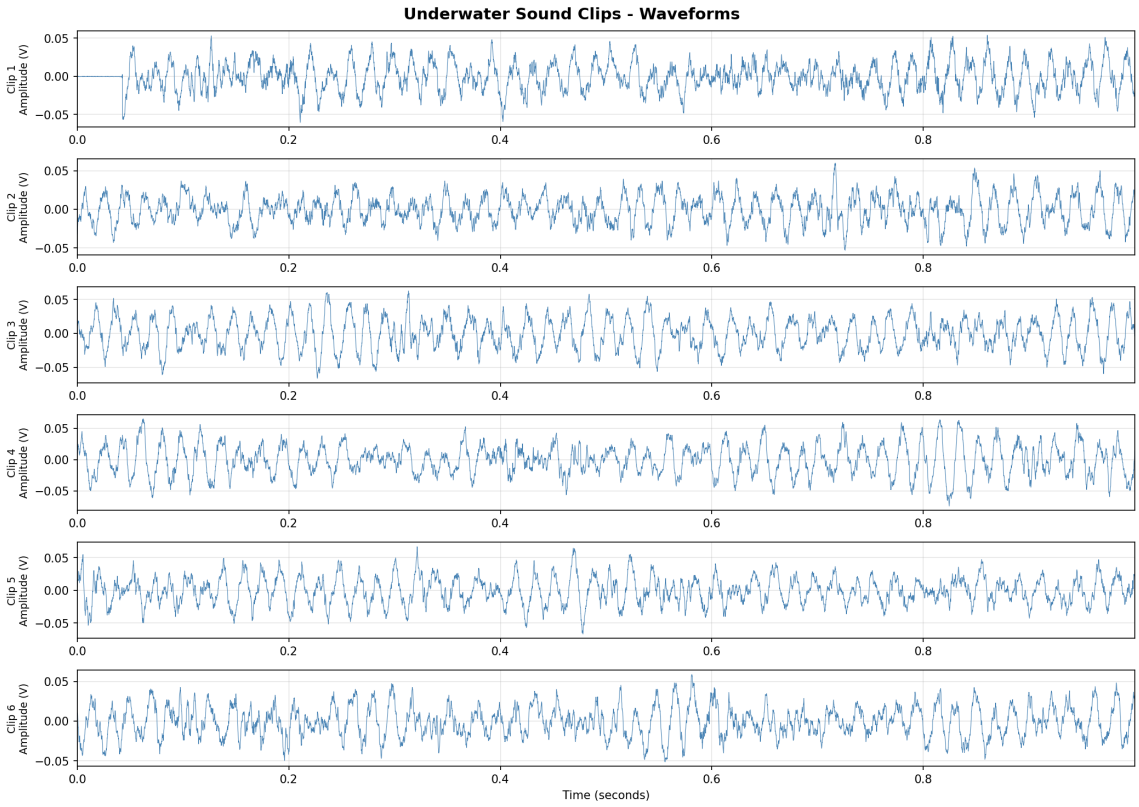

Figure 1: Time-domain waveforms of six underwater clips. X-axis: time (s), Y-axis: amplitude (V).

5.2 Spectrograms

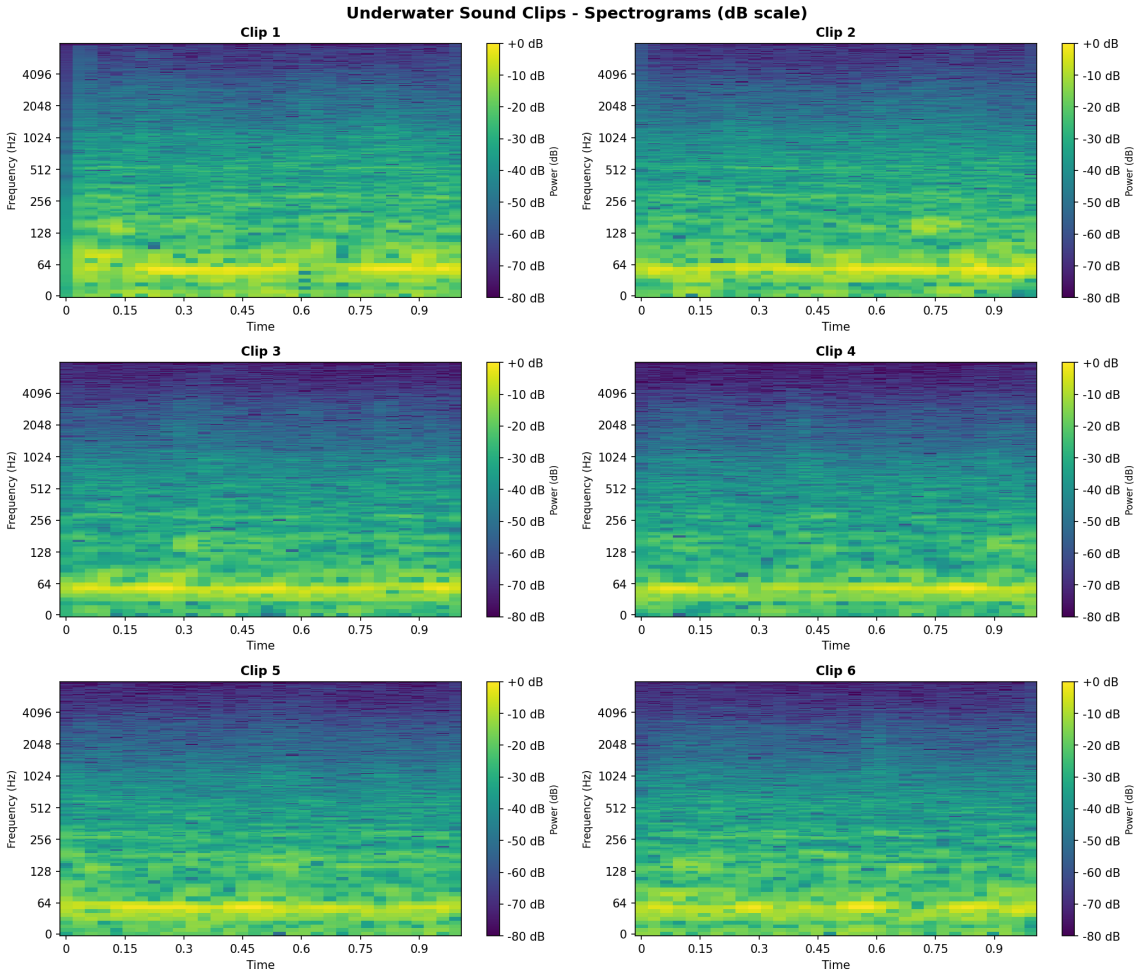

Figure 2: Spectrograms of six clips. Axes: time (s) and frequency (Hz, log scale). Color: power (dB).

5.3 Broadband Level Comparison

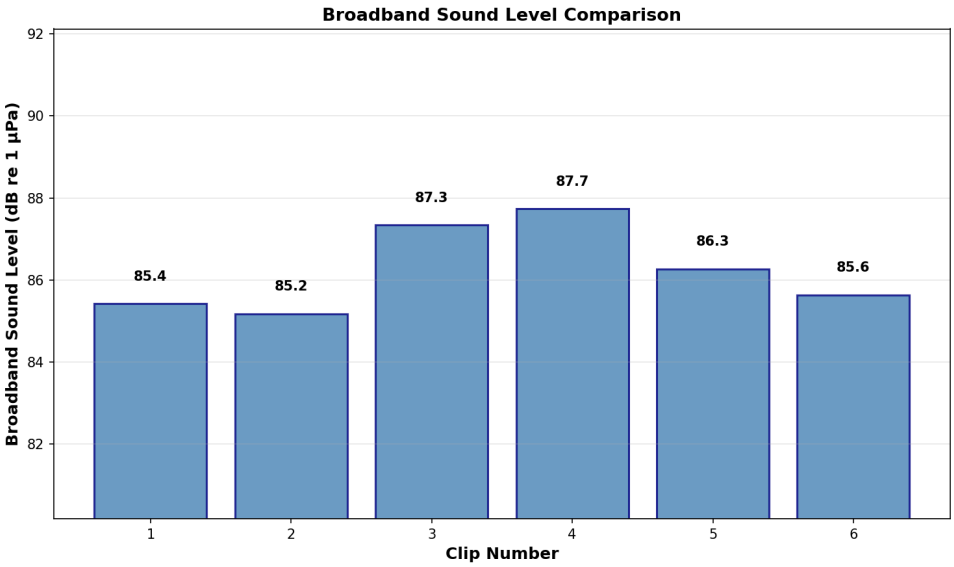

Figure 3: Broadband sound level comparison across clips. Unit: dB re 1 µPa.

5.4 One-Third Octave Band Levels

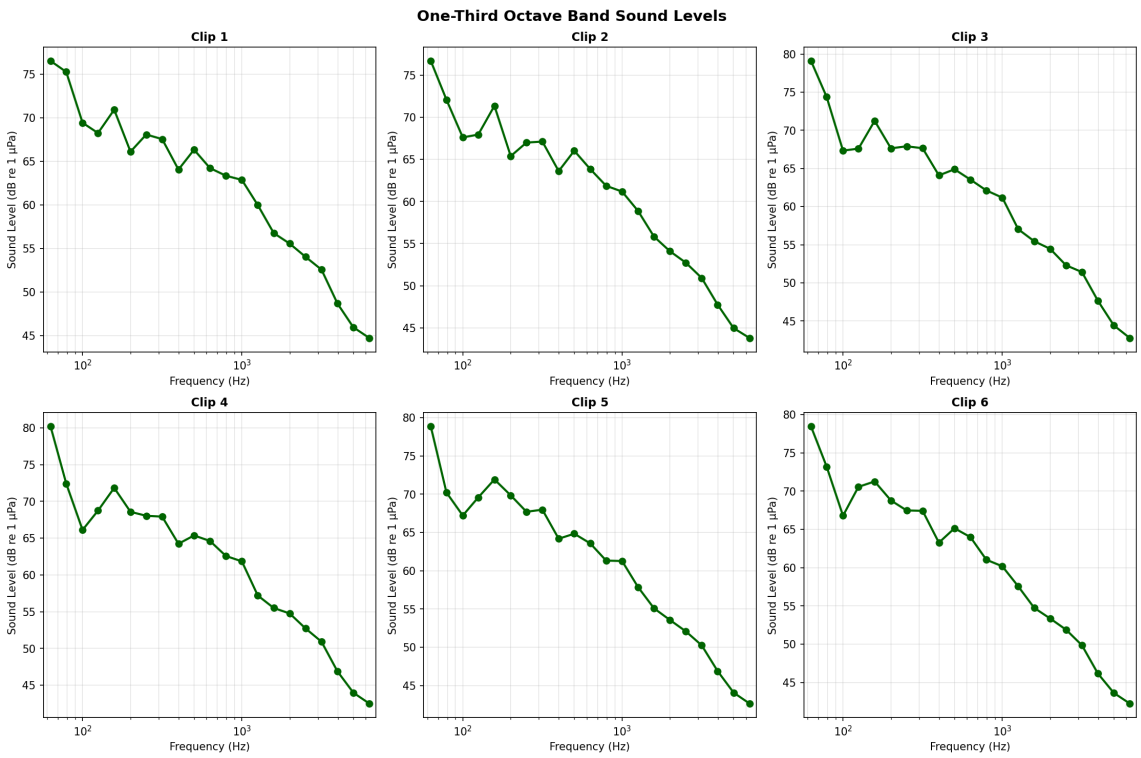

Figure 4: One-third octave band levels for each clip. X-axis: frequency (Hz), Y-axis: level (dB re 1 µPa).

5.5 Power Spectral Density

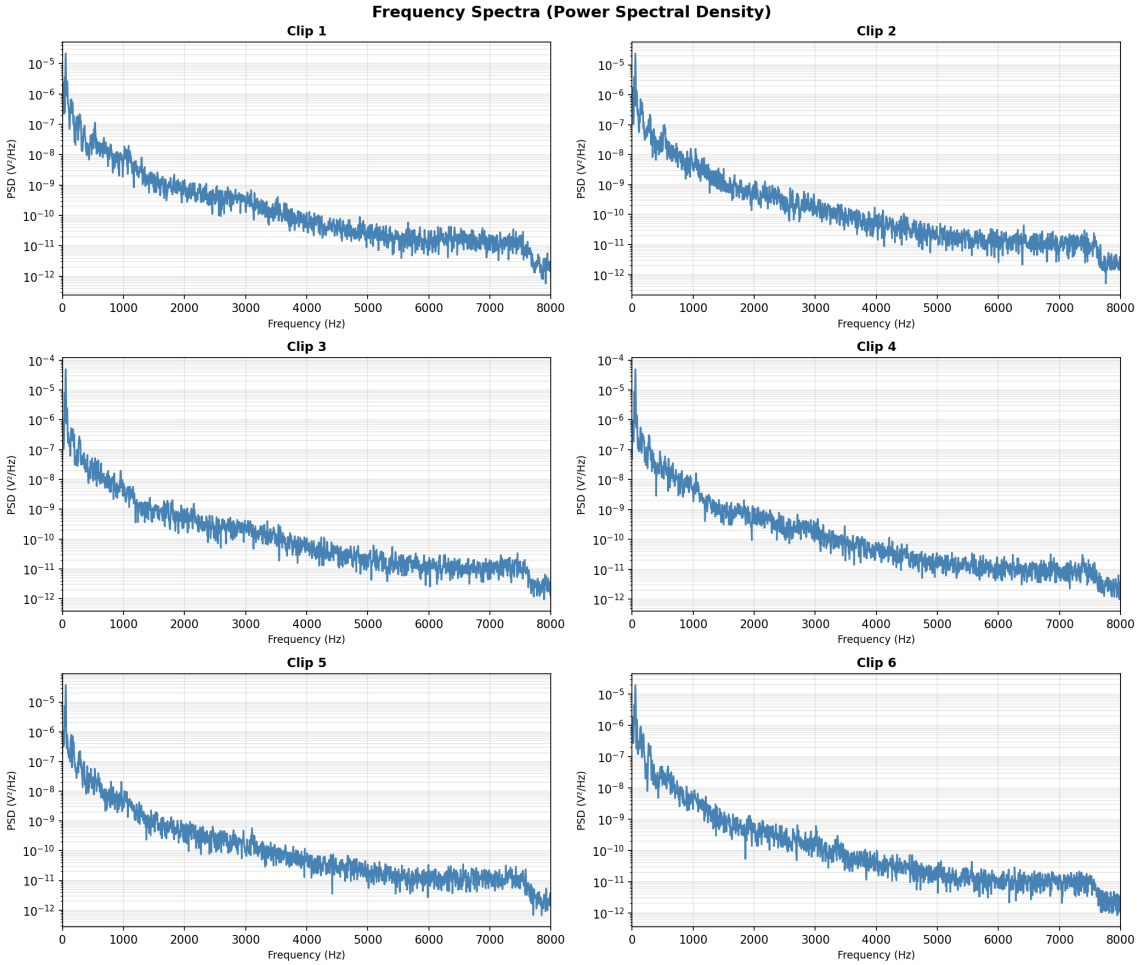

Figure 5: Welch PSD for six clips. X-axis: frequency (Hz), Y-axis: PSD (V^2/Hz).

6 Interesting Findings

6.1 Consistent Low-Frequency Signature

A strong and persistent spectral component is observed near 53 Hz in all clips. This repeatability indicates a stable source rather than random impulsive background noise.

6.2 Moderate and Stable Acoustic Intensity

Broadband levels remain within a narrow interval (85.2–87.7 dB re microPa), suggesting relatively steady source output over the analyzed window.

6.3 Energy Concentration at Low Bands

The one-third octave analysis peaks around 63 Hz, with limited high-frequency dominance. This pattern is typical of low-frequency machinery-like underwater sources.

6.4 Temporal Behavior

Crest factors between 2.84 and 3.30 indicate moderate impulsiveness, but not highly transient events. Combined with the stable dominant frequency, this supports a quasi-continuous source

process.

7 Interpretation and Source Hypotheses

Based on frequency concentration, stability, and level structure, the most plausible source category is low-frequency mechanical activity, such as vessel propulsion or industrial marine equipment. Biological contributions are possible but less consistent with the observed narrow low-frequency dominance and temporal steadiness.

8 Technical Notes and Reproducibility

- Main analysis script: `underwater_acoustic_analysis.py`
- Plot units are explicitly labeled on all axes and colorbars.
- Acoustic levels use underwater reference 1 µPa.
- Figures and values are regenerated automatically by running the script.

9 Conclusion

All requested deliverables are covered in this LaTeX report: methods, equations, unit-aware plots, quantitative tables, and notable findings from the dataset. The analyzed event in the Baltic Sea recording is dominated by a stable low-frequency component around 53 Hz, with moderate broadband levels and strong energy concentration in the 63 Hz octave region.